

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
Department of Information Technology

Subject Name: Data Structures And Algorithms
Subject Code: IT247

Semester: IV
Academic Year: 2018-19

Practical List
(Dec – Jun 2019)

Week - 1 : Array (2 hour) CO : 1, 2, 3 PO : 1 PEO : 1, 2	
1) Write a program to find duplicate elements in an array.	
2) Given an array of integers, segregate even and odd numbers in the array. All the even numbers should be present first, and then the odd numbers.	
Practical Assignment	
3) Write an efficient program to find the sum of contiguous subarray within a one-dimensional array of numbers which has the largest sum.	
4) Write a function rotate (arr[], d, n) that rotates arr[] of size n by d elements.	
Questions	
1) Describe advantages and disadvantages of Array.	
2) List out the applications of Array.	
Week – 2, 3, 4 : Searching & Sorting (6 hour) CO : 2, 3, 4 PO : 1, 7 PEO : 1, 2	
1) Let say you are a cashier in PQR Bank, people come to you when they want to withdraw money from their respective account. People have to come with their ADHAR CARD Photocopy. Prakash came and withdraw money from his account, of course by submitting adhar card photocopy, but on the second day he felt that submitted adhar card photocopy was not his, but adhar card no. he knew, which is (89891245). a) How will you find that particular adhar card? (Suppose total 15 people withdrew their money on that day). Assume there is no fix pattern in all adhar cards. b) How will you find that particular adhar card? (Suppose total 20 people withdrew their money on that day). Assume there is a pattern in all adhar cards, all are in ascending order (89891235 To 89891254). Develop this program using Iterative as well as Recursive approach.	
2) Write a program to implement Bubble sort and Selection sort.	
3) Write a program to implement Insertion sort to sort the cards in Playing Cards game.	
Practical Assignment	
4) Write a program to implement Merge sort.	
5) Find Kth largest and smallest element in an array.	
Questions	
1) Compare Linear Search and Binary Search.	
2) Compare performance of Bubble sort and Selection sort.	
Week – 5, 6 : Stack (4 hour) CO : 1, 2, 3 PO : 1, 2, 3, 7 PEO : 1, 2	
1) Write a menu driven program for Stack Operations using Array.(Push, Pop, Display)	
2) Write a program that enters valid Infix expression and convert it into Postfix expression.	
3) Write a program that enters valid Postfix/Prefix expression and evaluate the expression.	
Practical Assignment	
4) Write a program to print first n Fibonacci numbers using Recursion.	
5) Write a program to implement Tower of Hanoi Puzzle using Recursion. Tower of Hanoi is a mathematical puzzle where we have three stacks and n disks. The objective of the puzzle is to move the entire stack to another stack, obeying the following simple rules:	

1) Only one disk can be moved at a time. 2) Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack. 3) No disk may be placed on top of a smaller disk. 6) Write a program to check balanced parentheses in an expression using stack. Code can contain any brackets from the set []{}().

Questions

1) Differentiate Iterative Function and Recursive Function 2) List out the applications of Stack. 3) Why do we convert infix expression into postfix expression?
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Week – 7, 8 : Queue (4 hour)

CO : 1, 2, 3, 4 PO : 1, 2, 3, 7 PEO : 1, 2

1) We are developing software for a call center. When a client calls, his/her call should be stored until there is a free service representative to pick the call. Calls should be processed in the same order they are received. Select appropriate data structure to build call center software system. 2) Write a program to control Traffic Signal System. Switch on the traffic lights (Red, Yellow, and Green) one by one repeatedly as per the described time set.
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Practical Assignment

3) Write a program to implement Stack using Queue. 4) Write a program to implement Queue using Stack. 5) Write a Program to implement Double Ended Queue operations. (insert, delete, display)
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Questions

1) Compare Simple Queue and Circular Queue. 2) Compare Stack and Queue Data Structure 3) List out applications of Queue.
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Week –9, 10 : Linked List (4 hour)

CO : 1, 2, 3, 4 PO : 1, 2, 3, 7 PEO : 1, 2

1) Write code for creating a linked list, adding elements to the linked list at the tail, searching for an element with a particular key and also deleting an element with a particular key if it exists. 2) Write a program to implement Undo and Redo functionality of Text Editor. (Doubly Linked List)

Practical Assignment

3) Write a program to implement singly linked list and find following Count number of nodes, Find middle element, Reverse the singly linked list, get Nth node value 4) Write a program to implement dynamic Stack. (push, pop, display) 5) Write a program to implement dynamic Queue. (insert, delete, display) 6) Write a program to implement circular linked list. (insert, delete, display)

Questions

1) List out Advantages and Disadvantages of Doubly linked list over Singly Linked List. 2) Compare Array and Linked List data structure.

Week – 11, 12 : Tree (4 hour)

CO : 1, 2, 3 PO : 1, 2, 3, 7 PEO : 1, 2

1) Write a menu driven program to construct Binary Search Tree of your mobile phone number and apply different tree traversal techniques on BST. Find largest digit and smallest digit. Also search particular digit, If that digit is available in BST then prints its predecessor and successor.
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Practical Assignment

2) Write a program to implement level order and zigzag Tree traversal techniques. 3) Write a program to construct AVL tree.
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Questions

1) Draw BST for given Preorder and In-order traversal Preorder: 7,10,4,3,1,2,8,11 In-order: 4,10,3,1,7,11,8,2 2) List out applications of Tree.

Week – 13 : Graph (2 hour) CO : 1, 2,3 PO : 1, 2, 3,7 PEO : 1, 2
1) Write a program to implement BFS Graph traversal technique. 2) Write a program to implement DFS Graph traversal technique.
Practical Assignment
3) Write a program to find whether given graph contains cycle or not.
Questions
1) List out applications of Graph. 2) Compare BFS and DFS.

Week –14 : Hashing (2 hour) CO : 3 PO : 7 PEO : 2
1) In an array of 20 elements, arrange 15 different values, which are generated randomly between 1,00,000 to 9,99,999. Use hash function to generate key and linear probing to avoid collision. $H(x) = (x \text{ mod } 18) + 2$. Write a program to input and display the final values of array.
Practical Assignment
2) An array of 10 elements and hash function is $(x \text{ mod } 10)$. Handle Collisions using Separate Chaining. Keys: 75, 66, 42, 192, 91, 40, 49, 87, 67, 16, 417, 130, 372, 227 Write a program display the final values.
Questions
1) What is hashing? Discuss the importance of Hashing. 2) List out applications of Hashing.